ZIJIAN HUANG

Mobile: (+86) 188-1943-0147

Postdoctoral Researcher | Coastal Ecosystem Ecology

E-mail: huangzj79@mail.sysu.edu.cn

PROFESSIONAL EXPERIENCE

Post Doctoral Researcher, Ecology (Sun Yat-sen University) **Since Mar 2024**

Advisor: Professor Shixiao YU (School of Life Sciences)

Research Theme: How tree functional strategies shape mangrove canopy architecture, carbon storage, and ecosystem resilience.

RESEARCH INTERESTS

Coastal Wetland Ecology & Global Change: Ecosystem processes and responses to global change in mangroves and other coastal wetlands, with a focus on carbon cycling, ecosystem functioning, and biodiversity maintenance.

Plant Functional Strategies & Ecosystem Carbon Dynamics: Functional traits, whole-tree carbon allocation, and vegetation structural complexity as drivers of ecosystem functioning and resilience.

Landscape Ecology & Ecological Remote Sensing: Landscape pattern analysis using satellite imagery, three-dimensional vegetation structure characterization using terrestrial laser scanning (TLS), and multi-scale ecological monitoring and assessment.

EDUCATION

Doctorate in Natural Sciences in Ecology, Sun Yat-sen University **Sep 2017 – Dec 2023**

Supervisor: Professor Shixiao YU (School of Life Sciences)

Dissertation: Environmental Drivers and Effects on Reproductive and Leaf Growth Between Exotic and Native Mangroves Species

B. Sc. in Ecology, Sun Yat-sen University **Sep 2013 – Jun 2017**

Supervisor: Professor Shixiao YU (School of Life Sciences)

Thesis: Evaluating the Change of Urban Heat and Its Causes

PUBLICATIONS & MANUSCRIPTS

Accepted

- ✓ **Huang Z.,** Wu Y., Li Z., Zhu D., Yang Q., Yu S. (2026). A resilience dichotomy in mangrove forests: native resistance versus exotic compensation to climatic and tidal extremes. *Forest Ecosystems*. 16: 100441.
- ✓ **Huang Z.,** Yu S., Yang Q., Hu P. (2026). Mangrove protection drives decadal gains in post-typhoon productivity resilience through environmental adaptation. *Remote Sensing Applications: Society and Environment*. 41: 101944.
- ✓ **Huang Z.**, Li Z., Wang M., Zhu D., Yang Q., Yu S. (2023). Coastal nutrient enrichments facilitated reproductive output in exotic mangrove species over two decades. *Frontiers in Forests and Global Change*. 6: 1183432.
- ✓ Abulaiti Y., **Huang Z.**, Xie G., Zou X., Luo Q., Wang M., Yang Q., Hu P., Yu S. (2024). Comparison of resistance to pest infestation between native and exotic mangrove species. *Forest Ecology and Management*. 552: 121591.
- ✓ Pan J., **Huang Z.**, Yu S., Yang Q., Hu P. Tracking the decadal dynamics of native and exotic mangroves in an urban coastal wetland: an integrated OBIA-CNN approach. (2026). *Wetlands Ecology and Management*. 34: 41.
- ✓ Hu P., Xiang X., **Huang Z.**, Tan M., Yu S., Yang Q. (2024). Biomass dynamics in exotic and native mangrove species. *Chinese Journal of Applied Environmental Biology*. 30: 929-934. (In Chinese)
- ✓ Luo Q., Li Z., **Huang Z.**, Abulaiti Y., Yang Q., Yu S. (2023). Retrieval of Mangrove leaf area index and its response to typhoon based on WorldView-3 image. *Remote Sensing Applications: Society and Environment*. 30: 100931.
- ✓ Zhu D., Hui D., **Huang Z.**, Qiao X., Tong S., Wang M., Yang Q., Yu S. (2022). Comparative impact of light and neighbor effect on the growth of introduced species Sonneratia apetala and native mangrove species in China: implications for restoration. *Restoration Ecology*. 30: e13522.
- ✓ Wu Y., Ricklefs RE., **Huang Z.**, Zan Q., Yu S. (2018). Winter temperature structures mangrove species distributions and assemblage composition in China. *Global Ecology and Biogeography*. 27(12): 1492-1506.

- ✓ Hu L., Tan M., Luo Q., **Huang Z.**, Xiang X., Li B., Yu S., Wu Z., Yang Q., Hu P. (2024). Interannual changes of carbon storage in mangrove forests in Futian based on WorldView-3 remote sensing images. *Guihaia*. 44(8): 1403-1414.

Under review & in preparation

- ✓ Pan J., **Huang Z.**, Tan M., Zhou S., Qin F., Yu S., Yang Q., Ping H. Exotic mangroves compress niche differentiation among waterbird functional guilds: evidence from under -canopy monitoring of intertidal roost habitats. *Journal of Applied Ecology*. (Under review)
- ✓ Hu Y., Li B., **Huang Z.**, Yang Q., Hu P., Yu S. Scale-dependent habitat selection by *rhesus macaques* in a fragmented island reserve. *American Journal of Primatology*. (Under review)
- ✓ Abulaiti Y, **Huang Z.**, Li B., Yu S., Hu P., Yang Q. Spatial drivers of herbivore resistance divergence between native and exotic mangrove species. *Biological Invasions*. (Under review)
- ✓ **Huang Z.**, Qin F., Xie G., Yang Q., Hu P., Yu S. Canopy size and internal compactness underpin carbon–volume scaling across resource-use strategies in mangroves. *Ecology Letters*. (In submission).
- ✓ **Huang Z.**, Chen Y., Qin F., Tan M., Xiang X., Zhou S., Pan J., Hu Y., Cheng J., Ren Z., Yang K., Yan Q., Zheng F., Yu S. Biodiversity effects diverge across sequential components of mangrove stability. *Nature*. (In preparation).
- ✓ **Huang Z.**, Chen Y., Li B., Yu S. Climate-driven decoupling between growth and regeneration across an elevational gradient in subtropical forests. *Journal of Ecology*. (In preparation).

SELECTED RESEARCH PROJECTS

Vegetation Monitoring of the Futian Mangrove

2025 | Core Project Team Member

- ✓ Coordinated permanent-plot surveys, student activities, data digitization, and annual reporting.
- ✓ Collected and processed litterfall, TLS point-cloud, and leaf spectral data; implemented workflows for 3D tree reconstruction and spectral analysis.

Monitoring & 3D Mapping of Chebaling National Natural Reserve

2025 | Core Project Team Member

- ✓ Integrate field surveys with Google Earth Engine-derived remote-sensing data for multi-source ecological analysis.
- ✓ Conducted spatial interpolation and vegetation mapping in GIS to support reserve monitoring and spatial management.

Mycorrhizal Fungi-Mediated Forest Community Assembly

2022-2025 NSFC General | Participant

- ✓ Contributed to experimental design and refinement of hypotheses on forest community assembly and biodiversity patterns.
- ✓ Mentored students in experimental protocols, data quality control, and statistical analysis.

ACADEMIC SERVICES

Editorial Board Member: *Illustrated Guide to Plant Biodiversity in the Futian Mangrove Wetland*. (2026). Science Press, China.

Peer Reviewer: *International Journal of Applied Earth Observation and Geoinformation*, *Journal of Plant Ecology*, *Journal of Soil Science and Plant Nutrition*, *Forests*, *Remote Sensing*, *Wetland Science* (湿地科学), *Journal of Zhejiang A&F University* (浙江农林大学学报).

TECHNICAL SKILLS

Programming and Statistical Analysis: R & Python; Linux & Windows OS.

Remote Sensing & Spatial Analysis: Google Earth Engine (GEE), QGIS, and terrestrial laser scanning (TLS) data processing and 3D tree reconstruction.

Productivity & Media Software: Microsoft Office (Word, PowerPoint, Excel); Adobe Lightroom, Photoshop, Acrobat, and Premiere Pro.

Languages: Chinese – Native; English – Intermediate (IELTS Academic overall 7.0; CATTI Level 3 Written Translation).